What Is E-commerce?

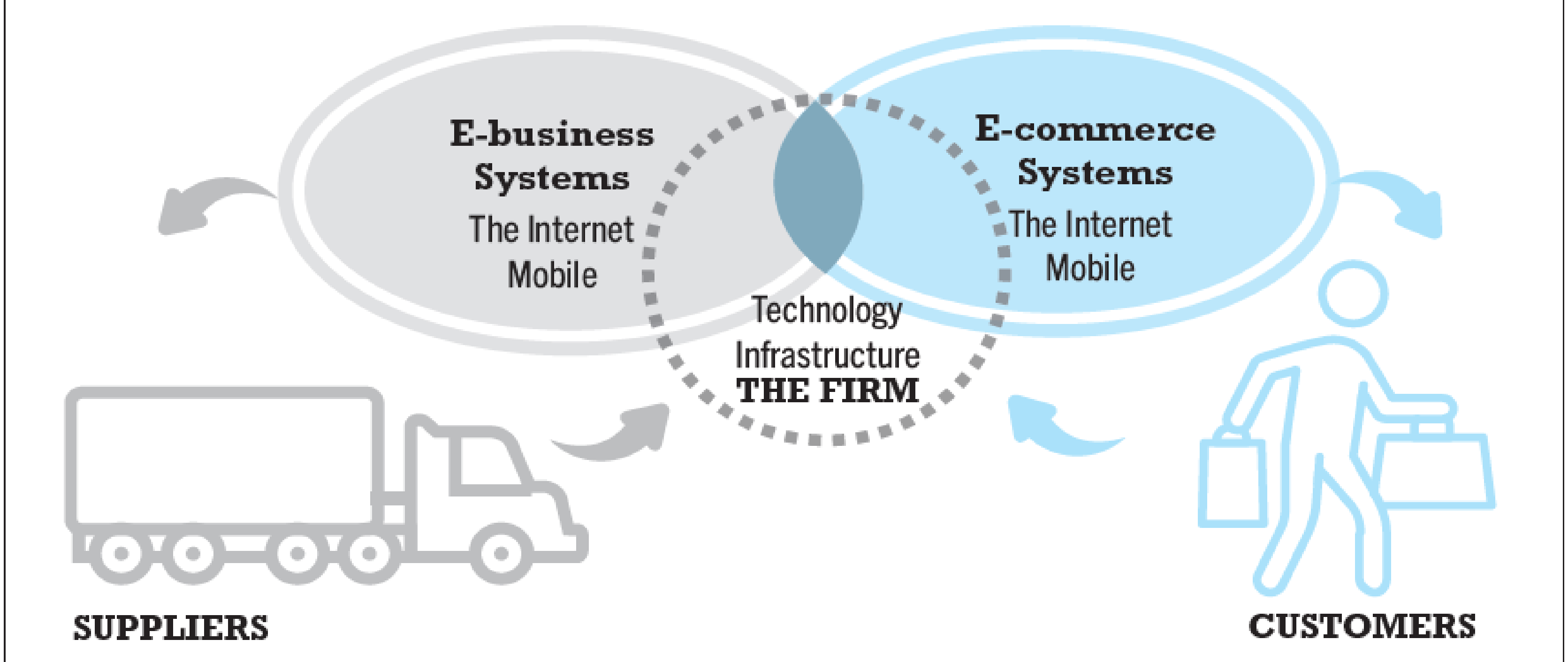
- Use of Internet and Web to transact business
- More formally:
  - Digitally enabled commercial transactions between and among organizations and individuals

E-commerce vs. E-business

- E-business:
  - Digital enabling of transactions and processes within a firm, involving information systems under firm’s control
  - Does not include commercial transactions involving an exchange of value across organizational boundaries

E-commerce vs. E-business

THE DIFFERENCE BETWEEN E-COMMERCE AND E-BUSINESS



E-commerce primarily involves transactions that cross firm boundaries. E-business primarily involves the application of digital technologies to business processes within the firm.

Why Study E-commerce?

- E-commerce technology is different, more powerful than previous technologies
- E-commerce brings fundamental changes to commerce
- Traditional commerce:
  - **Consumer as passive targets**
  - **Mass-marketing driven**
  - **Sales-force driven**
  - **Fixed prices**
  - **Information asymmetry**

Consumers as Passive targets

- **Meaning:**
  In traditional commerce, consumers are seen as passive recipients of products or marketing messages. Businesses push products toward them without much input or interaction from the consumers.
- **Illustration:**
  Think of a 1990s TV commercial for a shampoo. The brand airs the ad repeatedly during prime time. Viewers just watch it—they have no direct way to ask questions or compare products instantly. Their role is passive: watch, remember, and maybe buy.

Mass-marketing driven

- **Meaning:**
  Businesses use one-size-fits-all marketing strategies to reach as many people as possible with the same message, regardless of individual preferences.
- **Illustration:**
  A newspaper ad promoting a new car model is seen by thousands of readers, even though only a small fraction might be interested. The ad is not tailored to any specific group—just broadly targeted.

Sales-force driven

- **Meaning:**
  Salespeople play a central role in persuading customers to buy. The business often relies on personal selling more than product value or customer need.
- **Illustration:**
  In traditional retail or door-to-door selling, a salesperson visits a potential buyer, gives a pitch, and convinces them to make a purchase, often using persuasive techniques.

Fixed prices

- **Meaning:**
  Prices are usually fixed by the seller with little to no room for negotiation. Customers must accept the set price or leave.
- **Illustration:**
  In a shopping mall or department store, a pair of jeans has a fixed price tag. Unlike in some local markets, there's no haggling or personalized pricing.

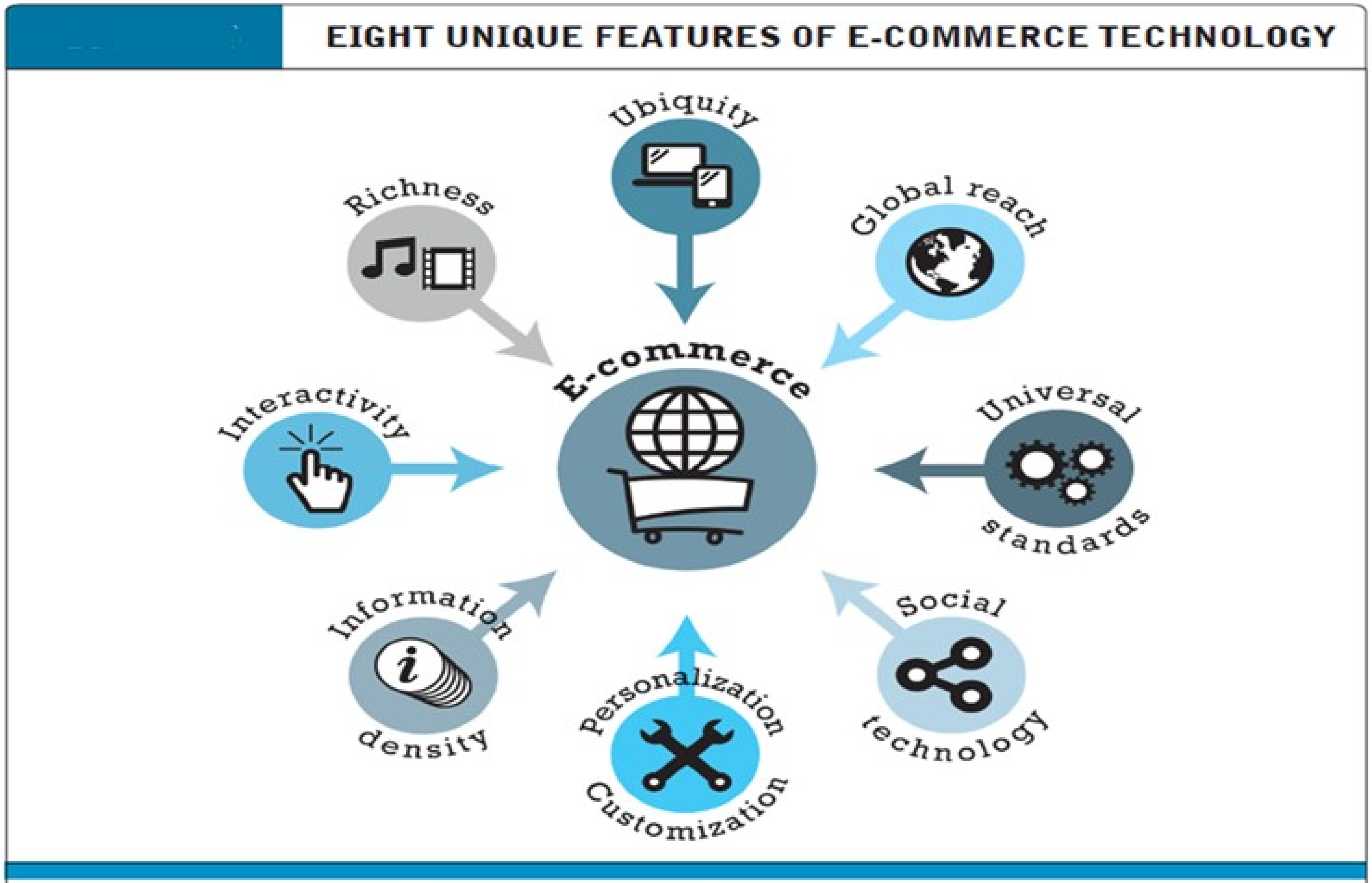
Information asymmetry

- **Meaning:**
  The seller knows much more about the product than the buyer. The buyer depends heavily on the seller's words and has limited ability to verify information.
- **Illustration:**
  When buying a second-hand car from a dealership in the 1980s, the buyer may not know if the car was in an accident. The seller has more information, and the buyer has little way to verify unless they bring an expert.

Eight Unique Features of E-commerce Technology

1. Ubiquity
2. Global reach
3. Universal standards
4. Information richness
5. Interactivity
6. Information density
7. Personalization/customization
8. Social technology

Eight Unique Features of E-commerce Technology



E-commerce technologies provide a number of unique features that have impacted the conduct of business.

Eight Unique Features of E-commerce Technology

BUSINESS SIGNIFICANCE OF THE EIGHT UNIQUE FEATURES OF E-COMMERCE TECHNOLOGY

E-COMMERCE TECHNOLOGY DIMENSION	BUSINESS SIGNIFICANCE
Ubiquity—E-commerce technology is available everywhere: at work, at home, and elsewhere via mobile devices, anytime.	The marketplace is extended beyond traditional boundaries and is removed from a temporal and geographic location. "Marketspace" is created; shopping can take place anywhere. Customer convenience is enhanced, and shopping costs are reduced.
Global reach—The technology reaches across national boundaries, around the earth.	Commerce is enabled across cultural and national boundaries seamlessly and without modification. "Marketspace" includes potentially billions of consumers and millions of businesses worldwide.
Universal standards—There is one set of technology standards.	There is a common, inexpensive, global technology foundation for businesses to use.
Richness—Video, audio, and text messages are possible.	Video, audio, and text marketing messages are integrated into a single marketing message and consuming experience.
Interactivity—The technology works through interaction with the user.	Consumers are engaged in a dialog that dynamically adjusts the experience to the individual and makes the consumer a co-participant in the process of delivering goods to the market.
Information density—The technology reduces information costs and raises quality.	Information processing, storage, and communication costs drop dramatically, while currency, accuracy, and timeliness improve greatly. Information becomes plentiful, cheap, and accurate.
Personalization/Customization—The technology allows personalized messages to be delivered to individuals as well as groups.	Enables personalization of marketing messages and customization of products and services based on individual characteristics.
Social technology—User-generated content and social networks.	Enables user content creation and distribution and supports development of social networks.

Ubiquity – *"Available everywhere, at any time"*

In traditional commerce, you had to **go to the store** during **specific hours** to shop. In e-commerce, the **store comes to you**, and it's **open 24/7**—wherever you are.

Supermarket vs. Online Grocery

Traditional: You need to visit a supermarket (e.g., Agora or Meena Bazar in Bangladesh), wait in line, and shop during their open hours.

E-commerce: You can order from **Chaldal.com** anytime—early morning or midnight—from your phone while lying in bed. Delivery comes to your door.

Physical Ticket Counter vs. Online Booking

Traditional: Buying a bus or train ticket meant standing in line at the station.

E-commerce: Use **Shohoz.com** or **Busbd.com** to book tickets online anytime—even while riding another bus!

Shopping Mall vs. E-commerce App

Traditional: You go to a shopping mall during opening hours.

E-commerce: Open the **Daraz app** from your living room or a tea stall and buy what you need in minutes.

Global Reach – ***"E-commerce can reach across national boundaries, to customers worldwide"***

Traditional commerce is often **limited to local or regional customers** due to physical, legal, or logistical barriers. But e-commerce breaks down those walls—you can sell or buy products from almost **any country in the world**.

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Local Shop vs. Etsy Seller

Traditional: A craftsman in Rajshahi makes jute handbags and sells them at the local market. Only local buyers can purchase them.

E-commerce: The same craftsman sells his bags on **Etsy.com**. Now, customers from **Canada, France, or Japan** can find, order, and pay online.

Local Author vs. E-book Publisher

Traditional: An author publishes a book and sells it through bookstores in Dhaka only.

E-commerce: Using **Amazon Kindle Direct Publishing**, the same author uploads an e-book. Readers in **London, Nairobi, and Toronto** can instantly download and read it.

Universal Standards – ***"E-commerce technologies use globally accepted technical standards that apply to all users and platforms"***

This means that **everyone uses the same rules, protocols, and formats** when engaging in e-commerce—whether they're in the USA, Bangladesh, or Japan. This standardization makes it **easy, fast, and cost-effective** to build and use e-commerce platforms.

Web Browser Compatibility

Traditional World: Think of trying to play a cassette from Bangladesh on a machine in Europe—they might not match formats.

E-commerce World: A website built with **HTML, CSS, and JavaScript** in Dhaka opens perfectly in **Chrome, Firefox, or Safari** anywhere in the world.

Online Shopping Carts

A developer in India creates a shopping cart feature using **HTTPS, cookies, and HTML forms**.

It works flawlessly for a buyer in Germany using a Mac, and another in Brazil using an Android phone—because the standards behind the code are universal.

Online Payment Gateways

Whether you're paying with **Visa, MasterCard, or PayPal**, the payment protocols (like **SSL encryption**, **HTTPS**, **JSON APIs**) are **standardized**.

This allows any e-commerce site to integrate payment systems **without custom setups for each country**.

Mobile E-commerce Apps

Developers build an e-commerce app using **React Native or Flutter**, and the app works on **both Android and iOS devices** globally.

Information Richness – ***"E-commerce can deliver detailed, multimedia-rich information about products and services"***

Compared to traditional commerce, where information is **limited and basic**, e-commerce can provide a **deep, engaging experience** using **text, images, videos, audio, interactive tools**, and more—all in one place.

Shoe Store vs. Online Product Page

Traditional: You walk into a shoe store and see shoes on the shelf. A salesperson gives a short pitch, but you can't see how the shoes are made or how they look on different people.

E-commerce: On **Nike.com**, you can see:

- Multiple images of the shoes from every angle
- A video showing how they perform during a run
- Customer reviews
- Size chart, material breakdown, FAQs

Phone Shop vs. Amazon Product Page

Traditional: You visit a mobile phone shop, hold the phone, and ask a few questions to the seller.

E-commerce: On **Amazon**, you can view:

- Detailed specifications
- Comparison with other phones
- User-generated Q&A
- Tutorial videos
- Star ratings and thousands of reviews

Restaurant Menu vs. Food Delivery App

Traditional: You go to a restaurant and read a plain text menu.

E-commerce: On **Foodpanda** or **Uber Eats**, you can:

- See pictures of each dish
- Watch short videos of how it's prepared
- Read customer ratings and reviews
- See allergy info and nutritional facts

Interactivity – *"E-commerce enables two-way communication between buyers and sellers, making the shopping experience more engaging, responsive, and personalized."*

Unlike traditional media (like TV or newspapers), which are **one-way communication**, e-commerce allows **two-way interactions**. This means users can **ask, respond, review, chat, customize**, and more — right on the platform.

Customer support at a shop vs. Chatbot on a website

Traditional: You walk into a shop and ask the staff questions about a product.

E-commerce: On **Daraz**, **Amazon**, or **AjkerDeal**, you use a **chatbot or live chat** to ask: "Is this available in black?" or "Can it be delivered to my area?" and get an instant reply.

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Street shopping vs. Interactive product selection

Traditional: You hold a product, ask the seller if there's a size or color, wait for their answer.

E-commerce: On **Nike.com**, you can **click to switch colors, see different sizes, read reviews**, or use tools to suggest your best fit.

Complaining in-store vs. Product reviews and seller responses

Traditional: You tell the manager in person that the product was bad.

E-commerce: On **Amazon**, you write a review: "The product broke in 3 days," and the **seller replies publicly** offering a refund or a solution.

Asking friends for advice vs. Community Q&A

Traditional: You ask your friend, "Is this phone good?" before buying.

E-commerce: On **Flipkart** or **AliExpress**, there's a **Q&A section** where users ask: "Is this camera waterproof?" and real buyers answer, "Yes, I tested it underwater."

Types of E-commerce

- *May be classified by market relationship or technology*
- Business-to-Consumer (B2C)
- Business-to-Business (B2B)
- Consumer-to-Consumer (C2C)
- Social e-commerce
- Mobile e-commerce (M-commerce)
- Local e-commerce

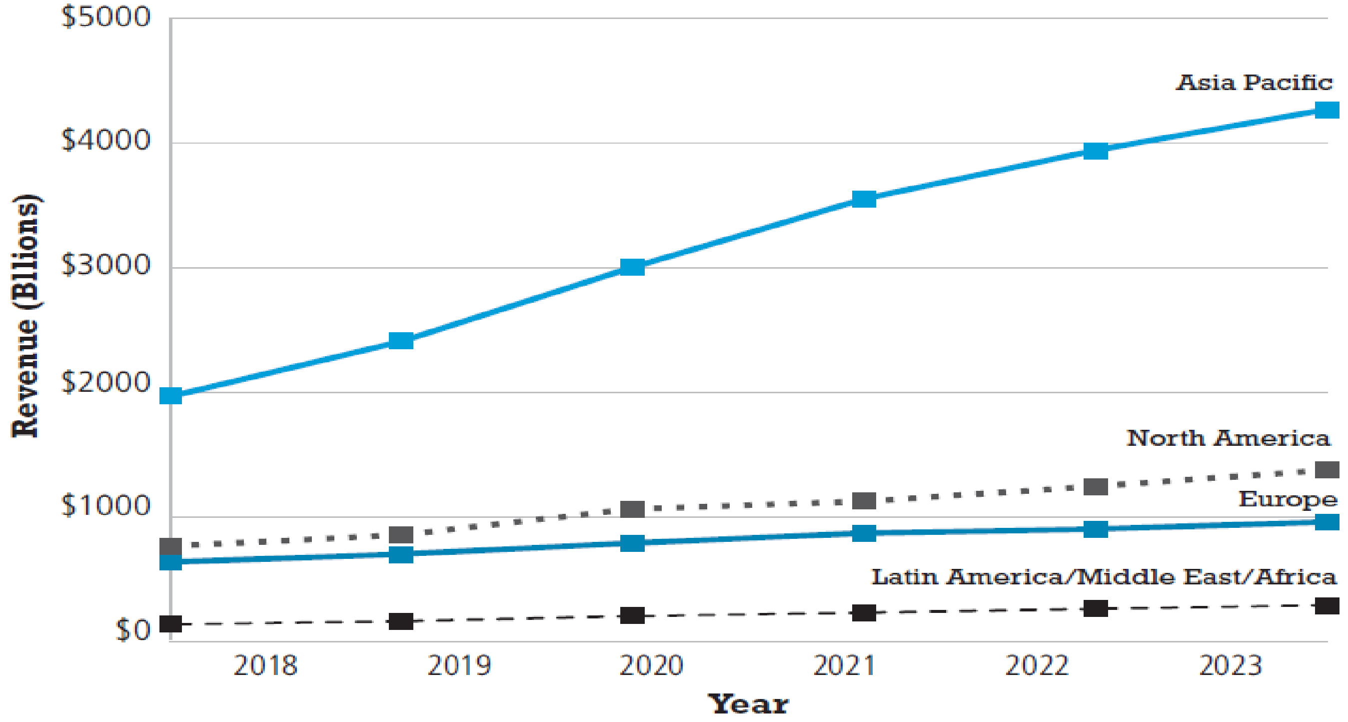
Types of E-commerce

MAJOR TYPES OF E-COMMERCE

TYPE OF E-COMMERCE	EXAMPLE
B2C—business-to-consumer	Amazon is a general merchandiser that sells consumer products to retail consumers.
B2B—business-to-business	Metalshub is an independent third-party marketplace that serves the metals industry.
C2C—consumer-to-consumer	Online platforms such as eBay, Taobao, and Gumtree enable consumers to sell goods directly to other consumers. Airbnb and Uber provide similar platforms for services such as room rental and transportation.
M-commerce—mobile e-commerce	Mobile devices such as tablet computers and smartphones can be used to conduct commercial transactions.
Social e-commerce	Facebook is both the leading social network and social e-commerce platform.
Local e-commerce	Groupon offers subscribers daily deals from local businesses in the form of Groupons, discount coupons that take effect once enough subscribers have agreed to purchase.

The Growth of B2C E-commerce

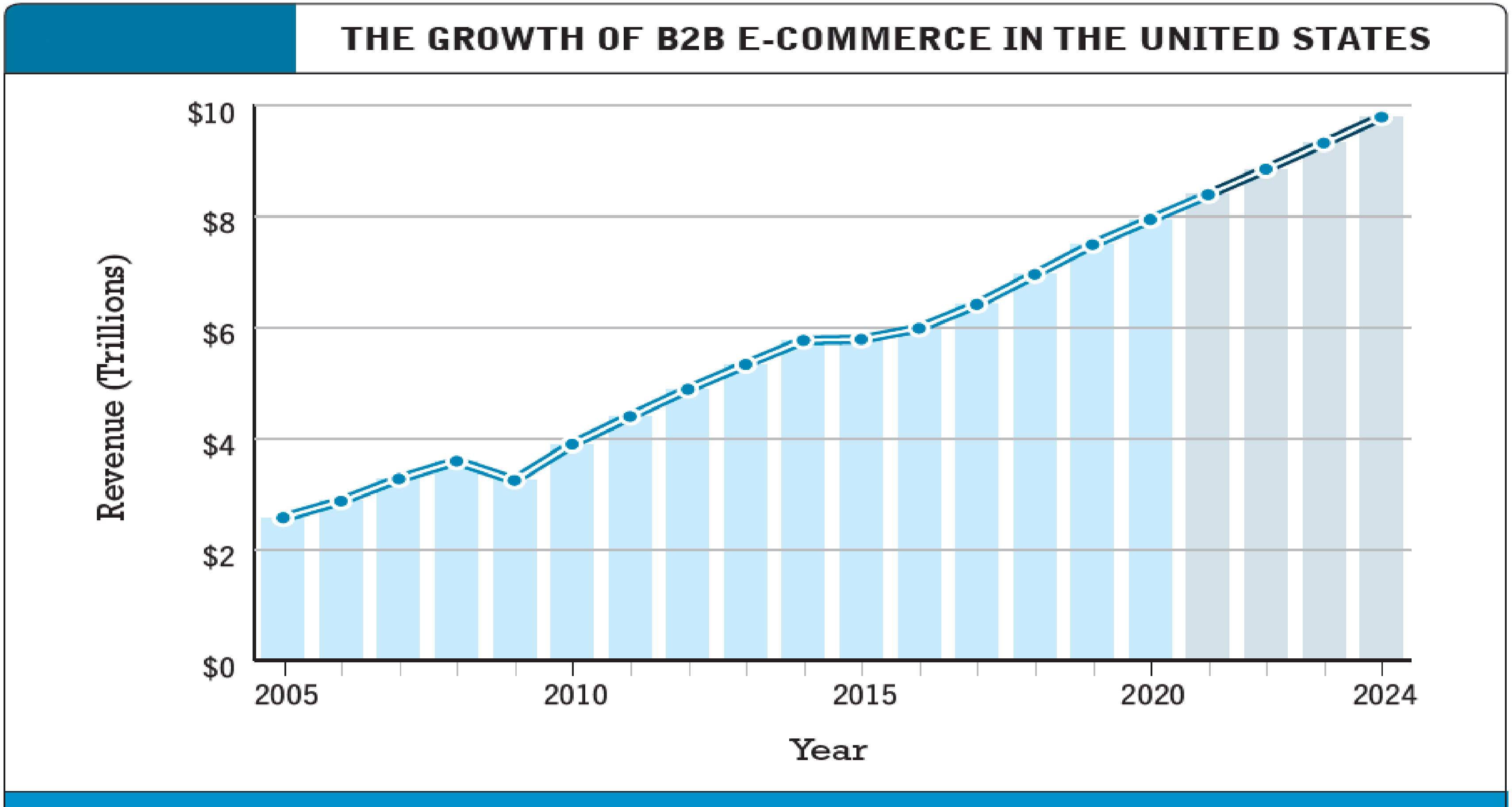
THE WORLDWIDE GROWTH OF B2C E-COMMERCE



B2C e-commerce revenues in all regions continued to grow throughout 2020. Retail e-commerce, the largest part of B2C e-commerce, surged throughout the world in 2020, in part due to the impact of the Covid-19 pandemic, offsetting the decline in digital travel revenues. The Asia-Pacific region generates the highest amount of B2C e-commerce revenue, followed by North America.

SOURCES: Based on data from eMarketer, Inc., 2020f, 2020g.

The Growth of B2B E-commerce



B2B e-commerce in the United States is over six times the size of B2C e-commerce. In 2024, U.S. B2B e-commerce is projected to be reach around $9.8 trillion. (Note: Estimates for 2020 through 2024 may be impacted by the Covid-19 pandemic.)

SOURCES: Based on data from eMarketer, Inc., 2020i; U.S. Census Bureau, 2019b; authors' estimates.

The Internet

- Worldwide network of computer networks built on common standards
- Created in late 1960s
- Services include the Web, e-mail, file transfers, and so on
- Can measure growth by number of Internet hosts with domain names

The Web

- Most popular Internet service
- Developed in early 1990s
- Provides access to Web pages
  - HTML documents that may include text, graphics, animations, music, videos
- Web content has grown exponentially
  - Google reports 30 trillion unique URLs; 120 billion Web pages indexed

The Mobile Platform

- Most recent development in Internet infrastructure
- Enables access to the Internet via wireless networks or cell-phone service
- Mobile devices include
  - Tablets
  - Smartphones
  - Ultra-lightweight laptops

Will Apps Make the Web Irrelevant?

Insight on Technology: Class Discussion

- What are the advantages and disadvantages of apps, compared with Web sites, for mobile users?
- What are the benefits of apps for content owners and creators?
- Will apps eventually make the Web irrelevant? Why or why not?

Origins and Growth of E-commerce

- Precursors:
  - Baxter Healthcare
  - Electronic Data Interchange (EDI)
  - French Minitel (1980s videotex system)
  - None had functionality of Internet
- 1995: Beginning of e-commerce
  - First sales of banner advertisements
- E-commerce fastest growing form of commerce in United States

E-commerce: A Brief History

- 1995–2000: Invention
  - Key concepts developed
  - Limited bandwidth and media
  - Euphoric visions of
    - Friction-free commerce
      - Lowered search costs, disintermediation, price transparency, elimination of unfair competitive advantage
    - First-mover advantages
      - Network profits
  - Dot-com crash of 2000

E-commerce: A Brief History (cont.)

- 2001–2006: Consolidation
  - Emphasis on business-driven approach
  - Traditional large firms expand presence
  - Start-up financing shrinks up
  - More complex products and services sold
  - Growth of search engine advertising
  - Business Web presences expand to include e-mail, display and search advertising, and limited community feedback features

E-commerce: A Brief History (cont.)

- 2007–Present: Reinvention
  - Rapid growth of:
    - Online social networks
    - Mobile platform
    - Local commerce
  - Entertainment content develops as source of revenues
  - Transformation of marketing
    - Coordinated marketing on social, mobile, local platforms
    - Analytic technologies

Start-up Boot Camp

- Why do you think investors today are still interested in investing in start-ups?
- What are the benefits of investing in a company that is a graduate of a Y Combinator boot camp?
- Is an incubator the best solution for start-ups to find funding? Why or why not?

Assessing E-commerce

- Many early visions not fulfilled
  - Friction-free commerce
    - Consumers less price sensitive
    - Considerable price dispersion
  - Perfect competition
    - Information asymmetries persist
  - Intermediaries have not disappeared
  - First mover advantages
    - Fast-followers often overtake first movers

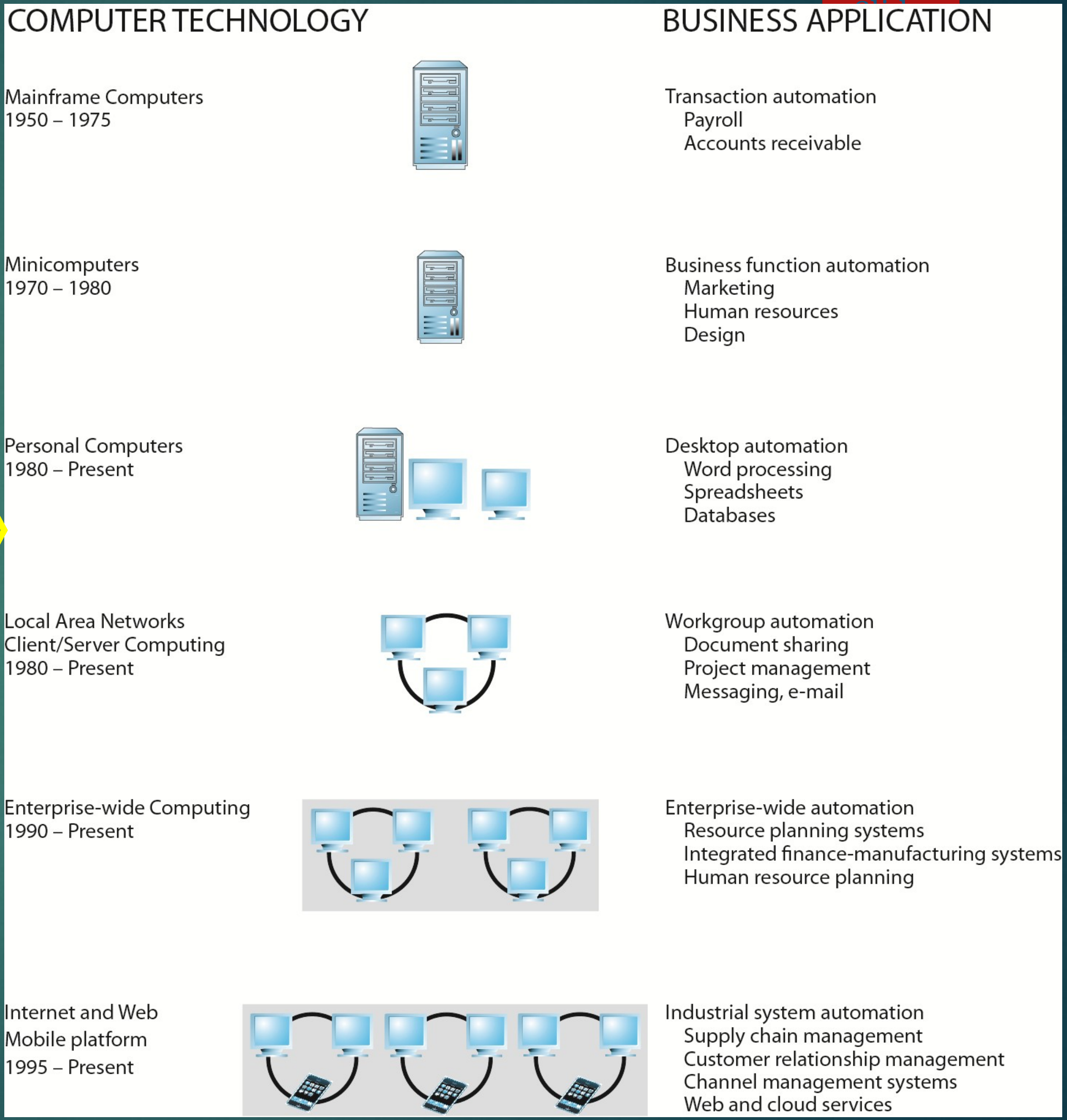
Predictions for the Future

- Technology will propagate through all commercial activity
- Large, traditional companies will continue to play dominant role, consolidating audiences
  - Start-up ventures can still attract large audiences in non-dominated arenas
- Integrated online/offline companies will experience more growth than purely online companies
- Additional factors:
  - Increased regulation and control
  - Cost of energy

Understanding E-commerce: Organizing Themes

- Technology:
  - Development and mastery of digital computing and communications technology
- Business:
  - New technologies present businesses with new ways of organizing production and transacting business
- Society:
  - Intellectual property, individual privacy, public welfare policy

The Internet and the Evolution of Corporate Computing



Facebook and the Age of Privacy

Insight on Society: Class Discussion

- Why are social network sites interested in collecting user information?
- What types of privacy invasion are described in the case? Which is the most privacy-invading, and why?
- Is e-commerce any different than traditional markets with respect to privacy? Don’t merchants always want to know their customer?
- How do you protect your privacy on the Web?

Academic Disciplines Concerned with E-commerce

- **Technical approach**
  - Computer science
  - Management science
  - Information systems
- **Behavioral approach**
  - Information systems
  - Economics
  - Marketing
  - Management
  - Finance/accounting
  - Sociology